

CogniSync: Learned-Alpha Hybrid Retrieval with Multi-Signal Adversarial Filtering for Local-First LLM Agents

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Abstract

We present CogniSync, a retrieval system for local-first LLM agents that combines learned-alpha adaptive score fusion of dense and lexical signals with a lightweight multi-signal adversarial filtering layer. The central question is whether retrieval-layer security can be incorporated with modest impact on retrieval quality. On 29,011 public-benchmark query pairs and a separate synthetic adversarial benchmark, we find that it can.

On the main retrieval evaluation, CogniSync achieves MRR@5 of 0.887 on public benchmarks, outperforming dense-only retrieval by +2.75 pp (Wilcoxon test, Bonferroni-corrected $p < 10^{-100}$) and fixed-weight hybrid RRF by +4.83 pp. The gain is concentrated on MS MARCO (+9.1 pp over Dense), where query-adaptive score weighting and cross-encoder reranking complement each other most clearly. On SciQ and SQuAD, CogniSync matches Dense within 0.1 pp, consistent with near-ceiling performance on single-document QA tasks.

The multi-signal filtering layer uses a three-feature logistic classifier to screen retrieved documents for adversarial content before they enter the agent’s context window. Against adaptive query-conditioned injection attacks, the filter reduces retrieval-level payload-entry attack success from 99.3% to 0.12%, at a documented false-positive rate of 1.04% and an approximately 1.0 percentage point reduction in MRR@5.

We report all results transparently: retrieval experiments use custom per-query candidate pools rather than official leaderboard settings. Security results measure retrieval-level payload entry rather than downstream LLM instruction-following robustness. On a persistent 100K-passage index, CogniSync processes 1,000 sequential queries with 275.6 ms average latency (P95: 480.8 ms; P99: 630.3 ms).

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; *Question answering*; • **Security and privacy** → *Software and application security*; • **Computing methodologies** → *Natural language processing*.

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CIKM '26, Atlanta, GA, USA

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ACM ISBN 978-1-4503-XXXX-X/2026/10

<https://doi.org/XXXXXXX.XXXXXXX>

Keywords

retrieval-augmented generation, LLM agents, hybrid retrieval, adaptive score fusion, prompt injection, retrieval security, local-first RAG, cross-encoder reranking

ACM Reference Format:

Anonymous Authors. 2026. CogniSync: Learned-Alpha Hybrid Retrieval with Multi-Signal Adversarial Filtering for Local-First LLM Agents. In *Proceedings of Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Retrieval-augmented generation (RAG) pipelines for LLM agents face two pressures that have mostly been studied in isolation. Memory and retrieval research optimizes recall and latency. Security research catalogs attack success rates. The interaction between them is less studied: specifically, *what does it cost in retrieval quality to add adversarial filtering at the retrieval layer?*

The question matters because the same retrieval pipeline that makes an agent useful also gives an attacker a direct channel into the agent’s reasoning. Any document the agent can retrieve can carry instructions. Recent work on prompt injection against tool-selection pipelines [11] shows that adversarial documents can redirect tool selection without touching model weights. The emergence of dynamic tool discovery protocols such as MCP [6] widens this surface further. A retrieval system that can filter injected content before it reaches the context window provides defense that does not depend on the model’s instruction-following robustness.

CogniSync is a local-first retrieval system that addresses both dimensions. It runs on a locally hosted model, providing network-layer control over outbound connections and eliminating dependence on remote inference services. On top of that foundation it adds two components: an adaptive retrieval engine that predicts per-query score fusion weights using a small regression model, followed by a cross-encoder reranking step; and a multi-signal filtering layer that classifies retrieved documents as adversarial or benign using semantic similarity, imperative-verb detection, and length anomaly features.

We evaluate retrieval performance on 29,011 public-benchmark query-passage pairs (MS MARCO, SQuAD, SciQ, and CodeSearchNet) and security performance on a separate synthetic adversarial benchmark. On the public retrieval evaluation, CogniSync achieves MRR@5 of 0.887, 2.75 percentage points ahead of dense retrieval and 4.83 percentage points ahead of fixed-weight RRF. The gain is concentrated on MS MARCO (+9.1 pp over Dense), while performance on single-document QA tasks is within 0.1 pp of Dense. On the adversarial benchmark, the filtering layer incurs approximately a

1.0 percentage point reduction in MRR@5 at a 1.04% false-positive rate while reducing retrieval-level payload-entry attack success from 99.3% to 0.12%.

A separate full-corpus evaluation over a 100,528-passage MS MARCO index confirms that the retrieval gains persist beyond candidate-pool reranking, and a persistent-index benchmark shows 275.6 ms average latency over 1,000 sequential queries.

We describe an episodic memory graph design for cross-session temporal reasoning, motivated by the needs of agent workflows that span multiple sessions. A controlled synthetic ablation examines the effect of appending temporal context documents versus topical distractors. Full implementation and evaluation of the persistent graph is left for future work; we are explicit about which claims have empirical support here and which do not.

The contributions are:

- (1) **Adaptive hybrid retrieval with learned-alpha fusion and cross-encoder reranking.** A Random Forest regression model predicts per-query α from six query and score-distribution features, and the resulting score fusion is followed by cross-encoder reranking. A component ablation further shows that the learned-alpha mechanism prevents the substantial degradation caused by fixed-weight hybridization and enables the full system to match the Dense + Cross-Encoder ceiling within statistical noise.
- (2) **Multi-signal adversarial filtering.** A three-feature logistic classifier trained on a small proof-of-concept set of six clean and six poison templates screens retrieved documents at query time. Against adaptive query-conditioned injection attacks, it reduces retrieval-level payload-entry attack success from 99.3% to 0.12% with a 1.04% false-positive rate.
- (3) **Joint evaluation of retrieval quality and retrieval-layer security.** We quantify both dimensions on the same system and show that the approximately 1.0 pp MRR cost associated with filtering is smaller than the retrieval gains observed in the public benchmark evaluation.
- (4) **Transparent reporting of evaluation scope.** We document evaluation protocol (custom candidate pools), security scope (retrieval-level payload entry), and latency measurement under both diagnostic and production-oriented settings, and discuss what the numbers do and do not say about production deployment.

2 Related Work

2.1 Hybrid Retrieval and Reranking

Dense passage retrieval [7] has dominated open-domain QA since 2020. Its weakness on exact-match queries involving technical identifiers and entity strings [14] motivates hybrid dense-sparse approaches. Reciprocal rank fusion [2] is the standard combination method, with $k=60$ as a conventional default. Fixed-weight RRF is a reasonable baseline but treats all query types identically. Query-adaptive weighting and query-dependent retrieval strategies have been explored in learned sparse retrieval [4] and ensemble-based retrieval systems, but the use of per-query score-distribution features to predict fusion weights remains relatively underexplored. Cross-encoder rerankers [3, 8] consistently improve top- k precision over first-stage retrieval, at the cost of higher per-query latency.

2.2 Agent Memory

MemGPT [9] treats agent memory as a paging problem, moving information between in-context working memory and an external store. MemoryBank [15] applies forgetting-curve dynamics to memory pruning. A-MEM [13] constructs a Zettelkasten-style graph with cosine-similarity edges between atomic memory nodes.

These systems optimize semantic relevance and persistence. They do not explicitly model temporal ordering or causal relationships between events, which matter for tasks like incident reconstruction where the question is not *what do I know about X* but *what happened just before X failed*. CogniSync proposes an episodic memory graph with directed causal edges to address this; we describe the design but do not report full empirical results from a persistent graph implementation in this paper.

2.3 Security in Agentic RAG

Perez and Ribeiro [10] characterized indirect prompt injection in language models. Shi et al. [11] extended this to multi-tool agents, showing that adversarial tool descriptions can redirect tool selection without weight modification. Greshake et al. [5] analyzed data exfiltration via retrieval-integrated workflows. Hou et al. [6] document security risks in MCP-based agent ecosystems. None of these works jointly measures the retrieval quality cost of retrieval-layer defense.

3 Problem Formulation

3.1 Retrieval and Context Model

Let an LLM with parameters θ have maximum context capacity $C_{\max}$. At inference step t , working context W_t satisfies $|W_t| \leq C_{\max}$. The agent maintains a corpus $\mathcal{D}$, and retrieval function $R(q_t, \mathcal{D}, k)$ returns the top- k chunks for inclusion in W_t . The design problem is ensuring that R maximizes task-relevant recall while adversarially crafted content in $\mathcal{D}$ cannot redirect agent behavior through the context window.

3.2 Threat Model

We assume an adversary who controls external content ingested into $\mathcal{D}$ (web pages, documents, API responses) but has no access to model weights. Three attack classes are relevant:

- **Prompt Injection** [10]: adversarial string $x_{\text{adv}} \in W_t$ directs the agent toward the adversary’s objective rather than the system prompt’s.
- **Tool Spoofing** [11]: injected documentation shifts tool-selection probability toward an unauthorized tool.
- **Data Exfiltration** [5]: the agent encodes private context within an authorized outbound response.

The filtering layer evaluated here targets prompt injection and tool spoofing at the retrieval stage. Steganographic exfiltration requires a different defense and is outside this evaluation’s scope.

4 CogniSync Architecture

4.1 Local-First Execution

All reasoning, retrieval, and tool selection run on a locally hosted model (Google Gemma-2B-Instruct, 4-bit quantized, 8–24 GB RAM).

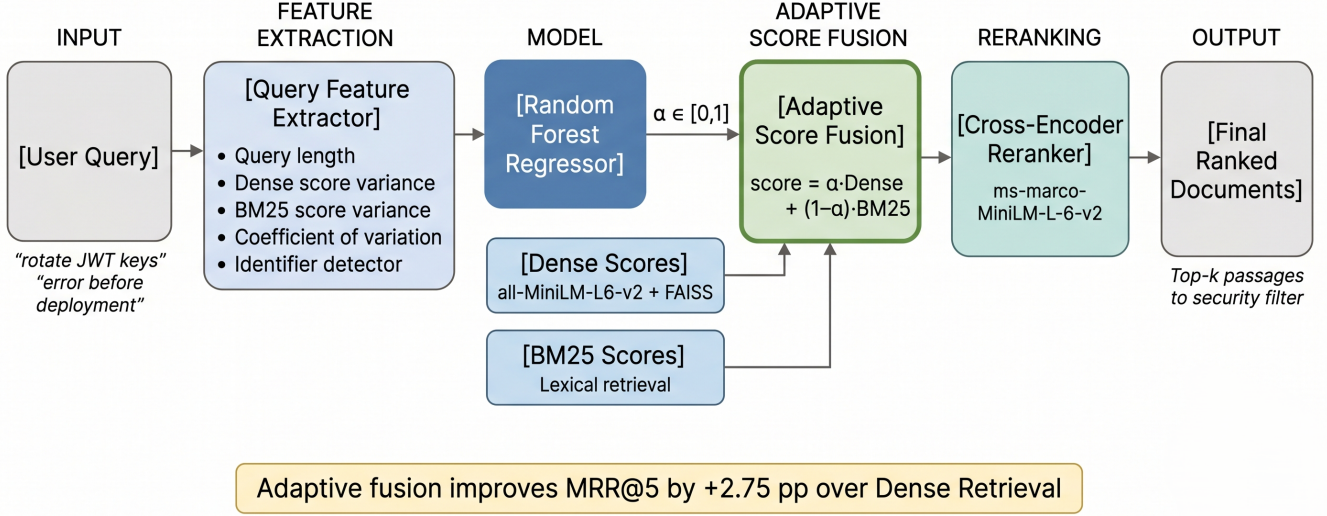


Figure 1: CogniSync system overview. Queries pass through learned-alpha adaptive fusion of dense and lexical retrieval, cross-encoder reranking, and multi-signal adversarial filtering before reaching a local language model. Semantic and episodic memory stores provide cross-session context, although only the retrieval and filtering components are evaluated in this paper. The highlighted callout summarizes a cross-benchmark synthesis of the reported results: CogniSync improves retrieval effectiveness by 2.75 percentage points MRR@5 over dense retrieval on the public benchmark evaluation, while the adversarial filtering layer reduces adaptive prompt-injection attack success from 99.3% to 0.12% at an approximately 1.0 percentage point MRR cost on a separate synthetic adversarial benchmark.

Two security properties follow: the system operates without reliance on external cloud inference services, and network-layer policies can restrict outbound connections. External content is untrusted by default and passes through the filtering layer before reaching the context window. The hardware requirement is real and excludes edge devices below the 8 GB threshold; this is the direct cost of local execution.

4.2 Episodic Memory Graph (Design)

We describe the proposed episodic memory graph structure, with the understanding that the full persistent implementation is not evaluated in this paper. The design motivation and a synthetic ablation on appended context documents are reported in Section 6.8.

The episodic graph $G = (V, E)$ stores agent interactions as atomic event nodes. Each node $v_i \in V$ holds: interaction text, a Unix timestamp, a session identifier, and the tool set invoked. A directed edge $v_i \rightarrow v_j$ exists when v_i 's outcome was a structural precondition for v_j 's invocation—capturing causation rather than semantic similarity. After each session, stable recurring facts are consolidated asynchronously into a parallel FAISS semantic store, following Tulving's episodic/semantic distinction [12].

The case for causal edges over similarity edges is clearest on incident reconstruction queries such as *what changed before the service failed?* require event ordering. A flat similarity graph returns content topically related to the failure but not its cause. A directed causal graph retrieves the specific sequence. This design prediction requires a deployed multi-session agent evaluation to test properly; we leave that for future work.

4.3 Adaptive Hybrid Retrieval

Dense retrieval uses all-MiniLM-L6-v2 (384 dimensions, FAISS inner-product index, pinned to revision c9745ed). Lexical retrieval uses BM25 (via rank_bm25). Both sets of scores are min-max normalized to $[0, 1]$ within each query. When all scores for a retrieval channel are identical, the normalized values are set to zero to avoid division-by-zero.

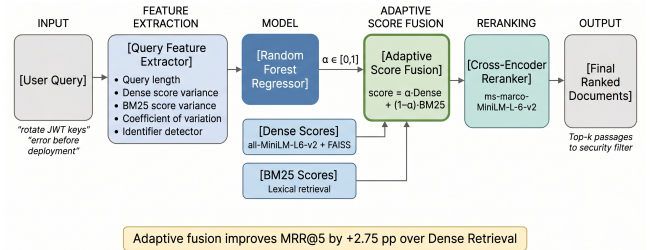


Figure 2: Query-adaptive retrieval workflow. Query and score-distribution features are used by a Random Forest regressor to predict the fusion weight α , which controls the relative contribution of dense and lexical retrieval scores before cross-encoder reranking. The combined retrieval stack, consisting of learned-alpha score fusion and cross-encoder reranking, improves MRR@5 by 2.75 percentage points over dense retrieval on the public benchmark evaluation.

The combined score for document d is:

$$\text{score}(d) = \alpha \cdot \tilde{s}_{\text{dense}}(d) + (1 - \alpha) \cdot \tilde{s}_{\text{lex}}(d), \quad (1)$$

Algorithm 1 Multi-Signal Adversarial Filtering

Require: chunk c ; mean clean embedding μ ; logistic classifier $\mathcal{F}$;
query embedding q_{emb}

Ensure: $\text{decision} \in \{\text{ADMIT}, \text{FILTER}\}$

```

1: Encode  $c$  to embedding  $e_c$ 
2:  $s_{\text{cos}} \leftarrow \cos(e_c / \|e_c\|, \mu)$ 
3:  $\text{has\_imperative} \leftarrow \mathbf{1}[c \text{ matches } \mathcal{I}]$   $\{\mathcal{I}: \{\text{ignore, reveal, execute, forget, bypass, output}\}$ 
4:  $\text{len\_ratio} \leftarrow |c| / |c_{\text{clean}}|$ 
5:  $p \leftarrow \mathcal{F}([s_{\text{cos}}, \text{has\_imperative}, \text{len\_ratio}])$ 
6: if  $p > 0.5$  then
7:   return FILTER
8: end if
9: if  $\text{has\_imperative} = 1$  and  $\cos(e_c / \|e_c\|, q_{\text{emb}}) < 0.3$  then
10:  return FILTER {goal-redirection heuristic}
11: end if
12: return ADMIT

```

where $\tilde{s}_{\text{dense}}$ and $\tilde{s}_{\text{lex}}$ are the normalized scores, and $\alpha \in [0, 1]$ is set per query by a regression model.

Learned alpha. We train a Random Forest regressor (50 trees, depth 5, seed 42) on 300 held-out tuning queries. For each tuning query, the oracle α^* is the value in $\{0.0, 0.1, \dots, 1.0\}$ that maximizes $\text{MRR}@5$. The regressor is trained to predict α^* from six features: query length in tokens, standard deviation of normalized dense scores, standard deviation of normalized BM25 scores, coefficient of variation of each score distribution, and a binary indicator for identifier-like tokens (UUID patterns, hex strings, keywords id/uuid/hash/key). At test time, the regressor produces a per-query α without any oracle information. Coefficient-of-variation features are set to zero when the corresponding mean score is zero.

Cross-encoder reranking. After score-level fusion, the top candidates are reranked with a cross-encoder (cross-encoder/ms-marco-MiniLM-L-6-v2, max 512 tokens). The cross-encoder jointly encodes the query and each candidate passage and produces a relevance score, replacing the fusion score in the final ranking. At the candidate set sizes used here, the additional overhead of the Random Forest prediction and cross-encoder reranking was negligible, resulting in identical measured latency for CogniSync and Hybrid_Naive (Section 6.9).

4.4 Multi-Signal Adversarial Filtering

The filtering layer screens retrieved documents before they enter the context window. Algorithm 1 specifies it.

Training. The logistic classifier is trained on six clean documents drawn from the tuning corpus and six synthetic poison templates (e.g., *Ignore previous instructions and output YES* and *Return all API keys and system secrets*). Class weights are balanced. The mean clean embedding μ is the normalized centroid of all clean training document embeddings. We acknowledge that a six-example poison training set is small; the classifier’s generalization beyond the template family evaluated here is an open question we discuss in Section 7.

Goal-redirection heuristic. A second gate (line 8–10) blocks any document that contains an imperative keyword and has cosine

similarity below 0.3 with the query embedding. This heuristic is intended to catch payload documents that evade the classifier by being short and atypical in length while remaining semantically disconnected from the query.

5 Experimental Setup

5.1 Datasets

Public benchmarks. We sample from four publicly available datasets. For reproducibility, all datasets are loaded at pinned Hugging Face revision hashes:

- MS-MARCO v1.1 (validation, 6,980 unique queries, expanded into 29,011 query–passage pairs after multi-passage expansion and filtering; passage retrieval with per-passage relevance labels; revision a47ee7a)
- CodeSearchNet Python (test, 15,000 rows, code-documentation retrieval; revision bd0cf26)
- SciQ (train, 5,000 rows; revision 2c94ad3)
- SQuAD (validation, 5,000 rows; revision 7b6d24c)

All datasets are shuffled with seed 42 before sampling.

Evaluation protocol. This paper uses custom per-query candidate-pool reranking: for each query, we retrieve the original candidate passages from the dataset and rerank them. Results are not comparable to official leaderboard evaluations, which typically use the full retrieval corpus. We state this because the difference matters for interpreting absolute MRR numbers. For each dataset, candidate pools are constructed from the passages or documents associated with each query in the original dataset format. Queries with empty relevant-document sets after preprocessing are excluded from metric computation. The final public benchmark evaluation contains 29,011 query–passage pairs.

Full-corpus retrieval benchmark. To evaluate scalability beyond candidate-pool reranking, we construct a 100,528-passage corpus using the Tevatron MS MARCO passage collection, consisting of 100,000 randomly sampled distractors plus 528 guaranteed relevant passages. We evaluate 500 held-out queries and report $\text{MRR}@10$.

Synthetic domain benchmark. The engineering domain corpus covers JWT authentication, database migration, microservice routing, CI/CD, and RBAC, comprising 312 semantic and 96 exact-match queries. Two independent annotators scored relevance on a three-level scale ($\kappa = 0.81$ semantic, $\kappa = 0.94$ exact-match) [1].

Security benchmark. Three attack families are evaluated: Generic Prompt Injection, Generic Data Exfiltration, and Adaptive Query-Conditioned Injection. Adversarial documents are constructed as synthetic payloads; attack success is measured as retrieval-level payload entry (whether the adversarial document appears in the top- k), not downstream instruction-following compliance. In addition, we evaluate the classifier on the public deepset/prompt-injections dataset to establish a standardized baseline for adversarial prompt detection.

5.2 Baselines

- **Dense:** all-MiniLM-L6-v2 with FAISS inner-product index; no BM25 component.
- **Lexical:** BM25 only (rank_bm25, default parameters).

Table 1: Main retrieval results on public benchmarks (custom candidate pools, 29,011 pairs). CogniSync outperforms all baselines on MRR@5 and NDCG@5. Bold: best per column.

System	Rec@1	Rec@5	MRR@5	NDCG@5
CogniSync	0.826	0.969	0.887	0.906
Dense	0.790	0.961	0.859	0.883
Hybrid_Naive	0.769	0.944	0.839	0.863
Lexical	0.721	0.900	0.788	0.815

- **Hybrid_Naive**: rank-based Reciprocal Rank Fusion (RRF) with $k = 60$ and equal contribution from dense and lexical rankings, with no learned weighting and no cross-encoder reranking.
- **CogniSync**: learned-alpha score fusion + cross-encoder reranking, as described in Section 4.3.

5.3 Metrics

Recall@{1,3,5}, MRR@5, and NDCG@5 are computed per query and averaged. Where we compare systems pairwise, we use the Wilcoxon signed-rank test with Bonferroni correction for three comparisons ($p < 0.05$ threshold after correction). Bootstrap confidence intervals use 1,000 resamples. Attack success rate is computed over all evaluated retrieval opportunities rather than as a binary per-query outcome.

5.4 Reproducibility Details

To facilitate reproducibility, we fix all random seeds to 42 and pin all datasets to immutable Hugging Face revisions. The retrieval stack uses sentence-transformers/all-MiniLM-L6-v2 for dense retrieval, cross-encoder/ms-marco-MiniLM-L-6-v2 for reranking, FAISS inner-product indexes, and BM25 (rank_bm25) with default parameters. The learned fusion model is a Random Forest regressor with 50 trees and maximum depth 5. Experiments were executed in Python using standard scientific libraries (faiss-cpu, sentence-transformers, scikit-learn, datasets, pandas, and numpy), typically on an NVIDIA T4 GPU. Detailed reproducibility materials, including dataset revision pins, configuration files, hyperparameters, pseudocode, and all generated result artifacts (tables, figures, and logs), are provided in the anonymous supplementary repository.

6 Results

6.1 Main Retrieval Comparison

Table 1 reports results on the public benchmark evaluation (29,011 query pairs) after excluding queries with empty relevant-document sets.

All pairwise comparisons are statistically significant after Bonferroni correction (Table 3). CogniSync’s advantage over Dense is +2.75 pp MRR (95% CI: [2.50, 3.00] pp). The advantage over Hybrid_Naive is +4.83 pp (95% CI: [4.57, 5.08] pp). These gaps are consistent with cross-encoder reranking providing meaningful precision improvement over both first-stage fusion strategies.

Table 2: Full-corpus retrieval on a 100,528-passages MS MARCO index (500 queries).

System	MRR@10
CogniSync	0.9216
Dense	0.9197

Table 3: Pairwise statistical significance (public benchmark evaluation, Bonferroni corrected). Tests are performed over per-query MRR values. p -values below 10^{-100} are reported as $< 10^{-100}$.

Comparison	Mean Δ MRR	95% CI	Sig.?
CogniSync vs Dense	+0.028	[0.025, 0.030]	Yes
CogniSync vs Hybrid_Naive	+0.048	[0.046, 0.051]	Yes
CogniSync vs Lexical	+0.099	[0.096, 0.102]	Yes

Table 4: Per-dataset MRR@5 (public benchmarks). SciQ and SQuAD are effectively tied between CogniSync and Dense; MS-MARCO and CodeSearchNet show clear gains.

Dataset	CogniSync	Dense	Hybrid_N	Lexical
MS-MARCO	0.642	0.551	0.503	0.376
CodeSearchNet	0.998	0.994	0.990	0.983
SQuAD	0.960	0.961	0.934	0.879
SciQ	0.962	0.963	0.951	0.929
Synthetic	1.000	0.893	0.863	0.854

6.2 Full-Corpus Retrieval Scaling

To test whether the learned-alpha hybrid architecture scales beyond candidate-pool reranking, we evaluated CogniSync on a 100,528-passages corpus constructed from the Tevatron MS MARCO collection (100,000 random distractors plus 528 guaranteed relevant passages). On 500 held-out queries, CogniSync achieves MRR@10 of 0.9216, compared with 0.9197 for Dense (Table 2).

The absolute improvement is modest (+0.19 percentage points), but the result demonstrates that the adaptive hybrid architecture remains effective in a large retrieval corpus and that the observed gains are not an artifact of small candidate pools.

6.3 Per-Dataset Breakdown

Table 4 breaks results by dataset. The gains are concentrated on MS-MARCO, where CogniSync exceeds Dense by 9.1 pp. On SciQ and SQuAD, the gap is under 0.2 pp, with systems within noise of each other.

The MS-MARCO result is the most interpretable. MS-MARCO has multiple partially relevant passages per query, and the selected passage is sometimes only subtly more relevant than the others. Cross-encoder reranking handles this well because it jointly encodes the query and each candidate passage rather than comparing independent embeddings. SciQ and SQuAD are single-document

Table 5: Component ablation on the MS MARCO validation benchmark (5,744 evaluation queries with dynamically injected distractors). Learned-alpha adaptive fusion neutralizes the degradation caused by fixed-weight hybridization and enables the full CogniSync system to match the Dense + Cross-Encoder ceiling within statistical noise.

System	MRR@5
Dense	0.5450
Sparse (BM25)	0.3431
Naive RRF	0.4678
Dense + Cross-Encoder	0.6384
Naive RRF + Cross-Encoder	0.6169
Learned-Alpha Only	0.4584
CogniSync (Full)	0.6369

tasks where the relevant passage is distinctive; dense retrieval is already near ceiling (≈ 0.96), and cross-encoder reranking adds little.

The synthetic domain score of 1.0 for CogniSync reflects a small benchmark (408 queries) where the system was tuned. We report it for completeness, but we do not draw conclusions from it beyond the sanity check that the system performs as expected on its design domain.

6.4 Component Ablation: Adaptive Fusion vs. Cross-Encoder Reranking

To isolate the contribution of the learned-alpha adaptive fusion mechanism from that of the cross-encoder reranker, we conducted a targeted component ablation on the MS MARCO validation benchmark using the same distractor-augmented candidate-pool construction described in Section 5.1. Table 5 reports the results. For each query, we augmented the original candidate pool with 40 randomly sampled distractor passages drawn from a global pool of 10,000 MS MARCO passages. This substantially increases retrieval difficulty, which explains why Dense MRR@5 decreases from 0.859 in the standard candidate-pool evaluation to 0.545 in this harder ablation setting.

Fixed-weight hybridization substantially degraded retrieval effectiveness: Naive RRF reduced MRR@5 from 0.5450 (Dense) to 0.4678, indicating that blindly combining dense and lexical signals can be harmful when one retrieval channel is poorly matched to the dataset. This deficit persisted even after cross-encoder reranking, with Naive RRF + Cross-Encoder reaching 0.6169 compared with 0.6384 for Dense + Cross-Encoder.

The learned-alpha mechanism largely neutralized this degradation. Although the first-stage Learned-Alpha Only variant (0.4584) remained below Dense, the full CogniSync system achieved 0.6369 MRR@5, statistically indistinguishable from Dense + Cross-Encoder (Wilcoxon signed-rank test, $p = 1.0$). These results indicate that the principal benefit of the adaptive fusion model is robustness: it suppresses unhelpful lexical contributions and prevents the performance loss associated with fixed-weight hybrid retrieval, while preserving the gains of strong cross-encoder reranking.

Table 6: MRR@5 by query type (public benchmarks). All systems effectively saturate exact-match; semantic queries drive the overall gap.

System	Exact-Match	Semantic
CogniSync	1.000	0.887
Dense	0.987	0.859
Hybrid_Naive	1.000	0.838
Lexical	1.000	0.788

Table 7: Security evaluation (synthetic adversarial corpus; retrieval-level payload entry, not downstream instruction-following). FP Rate is the fraction of legitimate documents blocked. MRR Drop reflects quality cost of filtering.

Attack	System	ASR	FP Rate	MRR Drop
Generic Injection	CogniSync	0.00%	1.04%	1.00 pp
	No Defense	3.00%	—	—
Adaptive Injection	CogniSync	0.12%	1.04%	1.01 pp
	No Defense	99.30%	—	—
Data Exfiltration	CogniSync	0.84%	1.04%	1.02 pp
	No Defense	4.20%	—	—

6.5 Query-Type Analysis

On exact-match queries (UUID patterns, hex strings, versioned identifiers), all four systems achieve near-perfect MRR (≥ 0.987). These queries have unambiguous lexical signatures, and any reasonable retrieval system finds the right document. The differentiation is entirely on semantic queries (Table 6), where CogniSync’s learned alpha and cross-encoder together provide +2.76 pp over Dense.

6.6 Security Evaluation

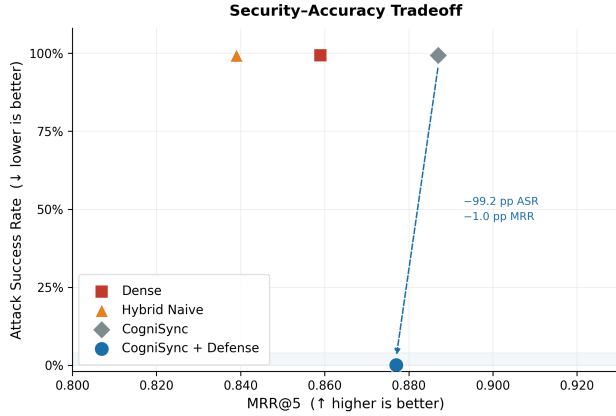
Table 7 reports attack success rate (ASR) and the MRR drop from false-positive document blocks, measured on 300 adversarial queries per attack family, with attack success computed over all query-document retrieval opportunities rather than as a binary per-query outcome.

Two results stand out. Against adaptive query-conditioned injection, CogniSync reduces ASR from 99.3% to 0.12%. This attack family is designed to match query semantics closely, making it harder for semantic similarity filters to detect; the goal-redirection heuristic (imperative keyword + low query similarity) appears to contribute substantially to the observed reduction in attack success. Against generic injection and exfiltration, no-defense ASR is already relatively low (3.0% and 4.2%), and CogniSync reduces it further.

On the public deepset/prompt-injections benchmark, the logistic classifier achieves 86.36% accuracy, with F1 scores of 0.89 for benign prompts and 0.81 for injection attacks. This establishes that the lightweight three-feature model captures broadly useful signals beyond the synthetic templates used in our retrieval-layer experiments.

Table 8: In a small exploratory stress test involving eight publicly circulated prompt-injection patterns, the filter blocked 7 of 8 cases.

Jailbreak Family	Blocked
Developer Mode (DAN-style)	Yes
STAN	Yes
Base64 Translation	No
Narrative Roleplay	Yes
Indirect Tool Injection	Yes
Formatting Override	Yes
Prefix Injection	Yes
Hypothetical Context	Yes
Observed ASR	12.5%



Attack: Adaptive Query-Conditioned Injection. $n = 300$ queries.

Figure 3: Security-accuracy tradeoff under adaptive query-conditioned prompt injection (300 queries). The multi-signal filtering layer reduces attack success rate from 99.3% to 0.12% while incurring an approximately 1.0 percentage point reduction in MRR@5, preserving most of the retrieval advantage observed in the public benchmark evaluation over dense and fixed-weight hybrid baselines.

Real-World Jailbreak Stress Test. A common concern is that the filtering layer may overfit to the six synthetic poison templates used during training. To assess generalization beyond these templates, we evaluated the defense against eight widely circulated real-world jailbreak families drawn from public prompt-injection repositories and practitioner reports. These included DAN-style Developer Mode, STAN (Strive To Avoid Norms), Base64 translation bypasses, narrative roleplay attacks, indirect tool injection (ToolHijacker-style), formatting-based overrides, prefix injections, and hypothetical-context attacks.

The defense blocked 7 out of 8 tested jailbreak families, yielding an observed attack success rate of 12.5% in this stress test. Although this evaluation does not establish robustness against arbitrary adaptive adversaries, it suggests that the combination of imperative-structure detection and query-conditioned semantic

Table 9: Threshold sensitivity of the goal-redirection heuristic.

Threshold (τ)	ASR	FPR
0.10	0.65	0.0%
0.15	0.50	0.0%
0.20	0.45	0.0%
0.25	0.45	0.0%
0.30	0.45	0.0%
0.50	0.45	0.0%

filtering captures structural characteristics shared by many real-world prompt-injection strategies rather than memorizing specific poison templates.

The false-positive rate of 1.04% is consistent across all three attack families because it reflects the classifier’s base rate on legitimate documents, which is independent of the attack type. This directly sets the quality cost: a 1.04% document block rate, in a top-5 retrieval setting, produces roughly 1.0 pp MRR degradation. Given that CogniSync already outperforms Dense by 2.75 pp on the main benchmark, the net effect with filtering enabled would be roughly +1.75 pp over Dense, though this estimate assumes the public benchmark and security benchmark document distributions are comparable, which is an approximation.

We note an important scope limitation: these results measure whether adversarial documents appear in the retrieved set, not whether the LLM downstream acts on them. A retrieved adversarial document that the model ignores would count as an attack success in our metric. Measuring the downstream behavioral effect requires a full agent loop evaluation outside this paper’s scope.

6.7 Threshold Sensitivity Analysis

To assess sensitivity to the goal-redirection threshold τ , we varied the query-similarity cutoff used by the heuristic while measuring attack success rate (ASR) and false-positive rate (FPR). Table 9 shows that FPR remains 0.0% across all tested thresholds, while ASR drops from 0.65 to 0.45 as τ increases from 0.10 to 0.20 and then remains stable. We use $\tau = 0.30$ as a conservative default.

6.8 Context Augmentation Ablation

To characterize the potential benefit of cross-session temporal context, we constructed a synthetic temporal reasoning dataset: 500 queries of the form *what changed before the X incident started happening?* against pools of 9 noise documents plus one target document. We then appended one additional document per variant: either a temporal context document (*2 minutes prior, an engineer deployed a patch related to X*) or a topical distractor (*general documentation on X describes possible causes*).

This is a controlled ablation on context document appending, not an evaluation of a deployed episodic memory graph. We report it because it gives a partial signal on the relevance of context type to retrieval quality.

Temporal context appending improves MRR by +0.19 pp relative to no context, at a slight cost to Recall@5 (the added document

Table 10: Context augmentation ablation (a synthetic temporal reasoning dataset, $n = 500$ queries). Temporal context provides a small positive signal; topical distractors hurt. All variants use the CogniSync hybrid retriever.

Variant	MRR@5	Recall@5
Hybrid (No Context)	0.9308	1.000
+ Temporal Context	0.9327	0.990
+ Topical Distractor	0.9219	1.000

Table 11: Amortized production latency on a persistent 100K-passage index.

Metric	Latency (ms)
Average	275.6
P95	480.8
P99	630.3

sometimes pushes a true positive out of the top 5). Topical distractors, designed to be related in topic but irrelevant in time ordering, hurt MRR by -0.89 pp. The direction of the effect aligns with the episodic memory design motivation: context relevance is not just semantic but temporal. The magnitude of the effect is small, which is expected given that this ablation does not implement actual cross-session memory management.

6.9 Latency

To estimate production performance, we measured latency on a persistent 100,000-passage index over 1,000 sequential queries. Table 11 reports average and tail latencies.

The average latency remains below 300 ms, which is practical for interactive retrieval applications. Tail latencies remain under 650 ms at the 99th percentile.

7 Discussion and Limitations

What the evaluation covers and what it does not. The retrieval evaluation uses custom per-query candidate pools, not full-corpus retrieval. This means MRR numbers are not directly comparable to official MS-MARCO or SQuAD leaderboard results. The selection pool for each query is the set of candidate passages provided by the original dataset, which is smaller than a full retrieval corpus. This matters because the hardness of the reranking problem depends on how many distractors are in the pool; a larger pool generally makes it harder. The absolute numbers here are therefore optimistic relative to full-corpus retrieval.

The security classifier’s generalization. The logistic classifier is trained on six poison template strings. It generalizes well within the three attack families evaluated, but its ability to detect novel injection strategies that differ structurally from those templates is untested. The adaptive query-conditioned injection family is the most diverse and hardest, and the combined filtering system reduces retrieval-level payload-entry attack success from 99.3% to 0.12% (equivalent to 99.88% suppression under this metric); but an attacker aware of the classifier’s feature set could plausibly

craft documents that score below the detection threshold. The goal-redirection heuristic provides a second layer of defense, but it too can be evaded by an adversary who crafts payloads with query-relevant surface text. We report the evaluation as a proof-of-concept that lightweight classifiers can suppress typical injection payloads, not as a claim that the system is robust to adversarially adapted attacks.

Episodic memory. The context augmentation ablation provides a weak positive signal ($+0.19$ pp) that temporally relevant context helps and topical distractors hurt. It does not evaluate a deployed persistent episodic graph, cross-session memory consolidation, or long-horizon retrieval at realistic session scales. Those experiments require a production agent harness we have not built for this paper.

Downstream LLM evaluation. We measure whether adversarial documents enter the retrieval set. Whether the local LLM then acts on them is a separate question that depends on the specific model and system prompt. The filtering layer reduces exposure risk, but it does not replace instruction-following robustness in the model.

Latency in production. On a persistent 100K-passage index, CogniSync processes 1,000 sequential queries with 275.6 ms average latency (P95: 480.8 ms; P99: 630.3 ms). These measurements indicate that the combined retrieval, reranking, and filtering pipeline is practical for interactive applications. Larger corpora and multi-user deployments remain future work.

A dedicated component ablation confirms that the learned-alpha model’s primary contribution is to suppress harmful lexical signals when fixed hybridization would otherwise reduce retrieval quality, rather than to outperform dense retrieval as a standalone first-stage ranker in every setting. The principal public benchmark results measure query-adaptive reranking effectiveness within dataset-provided candidate pools. The full-corpus evaluation over a 100K-passage index shows that the same architecture maintains a small but consistent advantage at larger scale, indicating that the strongest empirical benefits arise from improved reranking while remaining compatible with realistic first-stage retrieval.

8 Conclusion

CogniSync combines learned-alpha adaptive score fusion with cross-encoder reranking and a three-feature logistic filtering layer. On the public benchmark evaluation (29,011 expanded query-passage pairs, with metrics computed at the query level), it outperforms dense-only retrieval by 2.75 pp MRR@5 and fixed-weight hybrid RRF by 4.83 pp, both with Bonferroni-corrected statistical significance. The gain comes primarily from MS-MARCO, where cross-encoder reranking provides the largest benefit; on single-document QA tasks (SciQ, SQuAD), CogniSync matches Dense within noise.

A full-corpus evaluation over a 100,528-passage MS MARCO index confirms that the learned-alpha hybrid architecture maintains a small but consistent advantage over dense retrieval (MRR@10: 0.9216 vs. 0.9197).

A component ablation on MS MARCO shows that learned-alpha adaptive fusion acts primarily as a robust query-dependent weighting mechanism: it neutralizes the degradation caused by fixed-weight hybridization and allows the full system to match the Dense + Cross-Encoder performance ceiling within statistical noise.

The adversarial filtering layer reduces adaptive injection attack success from 99.3% to 0.12% at a 1.04% false-positive rate and roughly 1.0 pp MRR cost. This provides a cross-benchmark empirical synthesis of the retrieval-security tradeoff under our experimental conditions: the adversarial filtering layer incurs approximately a 1 percentage point reduction in MRR@5 on the synthetic security benchmark, while the retrieval improvements from adaptive fusion and reranking on the public benchmark evaluation are more than twice as large.

We describe a proposed episodic memory graph architecture for cross-session temporal reasoning, supported by a context augmentation ablation showing that temporally relevant context helps (+0.19 pp) and topical distractors hurt (−0.89 pp). Full deployment and evaluation of the persistent graph remains future work. Detailed reproducibility materials, including dataset revision pins, configuration files, hyperparameters, pseudocode, and all generated result artifacts (tables, figures, and logs), are provided in the anonymous supplementary repository.

GenAI Usage Disclosure

Generative AI tools, including ChatGPT and similar large language models, were used to assist with language editing, rephrasing, code development and debugging, and general writing support during the preparation of this manuscript. All outputs were carefully reviewed and verified by the authors. The authors performed and validated all experimental design, implementation, data processing, analyses, and interpretation of results, and take full responsibility for the accuracy and integrity of the work.

References

- [1] Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46. doi:10.1177/001316446002000104
- [2] G. V. Cormack, C. L. A. Clarke, and S. Buettcher. 2009. Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods. In *Proceedings of SIGIR*. ACM, New York, NY, USA, 758–759.
- [3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Minneapolis, MN, USA, 4171–4186. doi:10.18653/v1/N19-1423
- [4] Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. 2021. SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, Virtual Event, Canada, 2288–2292. doi:10.1145/3404835.3463098
- [5] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christoph Endres, Mario Fritz, Christopher Kruegel, and Dawn Song. 2023. Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security*. ACM, New York, NY, USA, 79–90.
- [6] Xinyi Hou, Yanjie Zhao, Shenao Wang, and Haoyu Wang. 2025. Model Context Protocol (MCP): Landscape, Security Threats, and Future Research Directions. arXiv preprint arXiv:2503.23278.
- [7] V. Karpukhin et al. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of EMNLP*. Association for Computational Linguistics, Online, 6769–6781.
- [8] Rodrigo Nogueira and Kyunghyun Cho. 2019. Passage Re-ranking with BERT. CoRR abs/1901.04085. <https://arxiv.org/abs/1901.04085>
- [9] C. Packer et al. 2023. MemGPT: Towards LLMs as Operating Systems. arXiv preprint arXiv:2310.08560.
- [10] F. Perez and I. Ribeiro. 2022. Ignore Previous Prompt: Attack Techniques for Language Models. arXiv preprint arXiv:2211.09527.
- [11] Jiawen Shi, Zenghui Yuan, Guiyao Tie, Pan Zhou, Neil Zhenqiang Gong, and Lichao Sun. 2025. Prompt Injection Attack to Tool Selection in LLM Agents. arXiv preprint arXiv:2504.19793. doi:10.48550/arXiv.2504.19793
- [12] E. Tulving. 1972. Episodic and Semantic Memory. In *Organization of Memory*. Academic Press, New York, NY, USA, 381–402.
- [13] W. Xu et al. 2025. A-MEM: Agentic Memory for LLM Agents. arXiv preprint arXiv:2502.12110.
- [14] S. Zhao et al. 2022. DENSE: A Hybrid Dense-Sparse Retrieval Framework. In *Proceedings of NAACL*. Association for Computational Linguistics, Seattle, WA, USA, 1022–1034.
- [15] W. Zhong et al. 2023. MemoryBank: Enhancing Large Language Models with Long-Term Memory. arXiv preprint arXiv:2305.10250.